

1. An object undergoing simple harmonic motion takes 0.25 s to travel from one point of zero velocity to the next such point. The distance between those points is 36 cm. Calculate the (i) period, (ii) frequency, and (iii) amplitude of the motion.

21

i) $T = 0.25 \times 2 = 0.5 \text{ s}$

ii) 2 Hz

iii) $A = \frac{0.36}{2} = 0.18 \text{ m}$

2. A block whose mass m is 680 gm is fastened to a spring whose spring constant k is 65 N/m. The block is pulled maximum 11 cm generating cosine form on a frictionless surface and released from rest at $t=0$. Find out (i) time period, (ii) angular frequency, (iii) phase constant, (iv) V (4 sec), $V_{\max}$, (v) a (4 sec), $a_{\max}$, (vi) displacement at $t=0$ sec, $t=7$ sec, and (vii) velocity at a displacement $x=0.11 \text{ m}$, $x=0.04 \text{ m}$.

21

i) $k = \omega^2 m$
 $\Rightarrow \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{65}{0.680}} = 9.77 \text{ rad/s}$
 $T = \frac{2\pi}{\omega} = \frac{2\pi}{9.77} = 0.6426 \text{ sec}$

$k = 65 \text{ N/m}$
 $m = 680 \text{ g}$
 $= 0.680$
 $A = 0.11 \text{ m}$

ii) $\omega = 9.77 \text{ rad/s}$

iii) phase constant $\phi = 0$

iv) $V_{t=4} = A\omega \sin(\omega t)$
 $= 0.11 \times 9.77 \times \sin(9.77 \times 4)$
 $= 0.677 \text{ m/s}$

$V_{\max} = \omega A$
 $= 9.77 \times 0.11 = 1.07 \text{ m/s}$

v) $a_{t=4} = \omega^2 A \cos(\omega t)$
 $= (9.77)^2 \times 0.11 \times \cos(9.77 \times 4)$
 $= 8.44 \text{ m/s}^2$

$a_{\max} = \omega^2 A = (9.77)^2 \times 0.11 = 10.49 \text{ m/s}^2$

vi)

$x = A \cos(\omega t)$

$x_{t=0} = A = 0.11 \text{ m}$

$x_{t=7} = (0.11) \cos(9.77 \times 7) = 0.0405 \text{ m}$

vii)

$V = \omega \sqrt{A^2 - x^2}$

$V_{x=0} = 9.77 \sqrt{(0.11)^2 - (0.11)^2} = 0 \text{ m/s}$

$V_{x=0.04} = 9.77 \sqrt{(0.11)^2 - (0.04)^2} = 1.0011 \text{ m/s}$

3. A 0.12 kg body undergoes simple harmonic motion of amplitude 8.5 cm and period 0.20 s. (i) What is the magnitude of the maximum force acting on it? (ii) If the oscillations are produced by a spring, what is the spring constant?

31

i) $F_{\max} = m a_{\max}$
 $= 0.12 \times 83.89$
 $= 10.06 \text{ N}$

$a_{\max} = \omega^2 A$
 $= \left(\frac{2\pi}{0.20}\right)^2 \times 0.085$
 $= 83.89 \text{ m/s}^2$

ii)

$k = \omega^2 m$
 $= \left(\frac{2\pi}{0.20}\right)^2 \times 0.12$
 $= 118.49 \text{ N/m}$

4. In an electric shaver, the blade moves back and forth over a distance of 2.0mm in simple harmonic motion, with a frequency 120Hz. Find (i) the amplitude, (ii) the maximum blade speed and (iii) the magnitude of the maximum acceleration of blade.

$$\begin{aligned}
 &4) \text{ i) } A = \frac{2\text{mm}}{2} \\
 &\quad = 10^{-3}\text{m} \\
 &\text{ii) } v_{\max} = \omega A \\
 &\quad = 2\pi f A = 2\pi \times 120 \times 10^{-3} \\
 &\quad = 0.7539\text{ m/s} \\
 &\text{iii) } a_{\max} = \omega^2 A = (2\pi f)^2 A = (2\pi \times 120)^2 \times 10^{-3} \\
 &\quad = 568.48\text{ m/s}^2
 \end{aligned}$$

5. An oscillator consists of a block attached to a spring ($k = 400\text{ N/m}$). At some time t , the position (measured from the system's equilibrium location), velocity, and acceleration of the block are $x = 0.100\text{ m}$, $v = -13.6\text{ m/s}$, and $a = -123\text{ m/s}^2$. Calculate (i) the frequency of oscillation, (ii) the mass of the block, and (iii) the amplitude of the motion.

$$\begin{aligned}
 &5) \text{ i) } a = \omega^2 x \\
 &\quad \Rightarrow \omega = \sqrt{\frac{a}{x}} = \sqrt{\frac{-123}{0.1}} = 35.07\text{ rad/s} \\
 &\quad \therefore \omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi} = \frac{35.07}{2\pi} = 5.58\text{ Hz} \\
 &\text{ii) } \omega = \sqrt{\frac{k}{m}} \Rightarrow m = \frac{k}{\omega^2} = \frac{400}{(35.07)^2} = 0.325\text{ kg} \\
 &\text{iii) } \frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \\
 &\quad \Rightarrow kA^2 = mv^2 + kx^2 \\
 &\quad \Rightarrow A = \sqrt{\frac{mv^2}{k} + x^2} \\
 &\quad = \sqrt{\frac{0.325 \times (-13.6)^2}{400} + (0.1)^2} \\
 &\quad = 0.4\text{ m}
 \end{aligned}$$

6. A body of mass 25gm is attached with a spring of spring constant 400dyns/cm. The body is displaced by 10cm from its equilibrium position and released. Then the body executes simple harmonic motion. Calculate (i) the time period, (ii) frequency, (iii) angular frequency and (iv) maximum velocity.

$$\begin{aligned}
 &6) \text{ i) } \omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{4}{0.025}} = 4\text{ rad/s} \\
 &\quad T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57\text{ sec} \\
 &\text{ii) } f = \frac{1}{T} = \frac{1}{1.57} = 0.64\text{ Hz} \\
 &\text{iii) } \omega = 4\text{ rad/s} \\
 &\text{iv) } v_{\max} = \omega A = 4 \times 0.1 = 0.4\text{ m/s}
 \end{aligned}
 \quad \left\{ \begin{array}{l} m = 0.025\text{ kg} \\ k = 400\text{ dynes/cm} \\ \quad = 0.4\text{ N/m} \\ A = 0.1\text{ m} \end{array} \right.$$

7. A hydrogen atom has a mass of $1.68 \times 10^{-27} \text{ kg}$, when it attach to a certain massive molecule, it oscillate as classical oscillator with frequency of 10^{14} Hz and with amplitude of 10^{-10} m . Calculate force acting on the hydrogen atom.

$$\begin{aligned} \text{7)} \quad F &= ma \\ &= m \omega^2 A \\ &= 1.68 \times 10^{-27} \times (2\pi \times 10^{14})^2 \times 10^{-10} \\ &= 6.633 \times 10^{-8} \text{ N} \end{aligned}$$

8. A body executes SHM such that its velocity at mean position is 1 m/s and acceleration at one extremity is 1.57 m/s^2 . Calculate time period of oscillation.

$$\begin{aligned} \text{8)} \quad v_{\text{max}} &= \omega A = 1 \text{ m/s} \\ a_{\text{max}} &= \omega^2 A = 1.57 \text{ m/s}^2 \\ \therefore \omega^2 \frac{1}{\omega} &= 1.57 \Rightarrow \omega = 1.57 \text{ Hz} \\ T &= 4.0020 \text{ sec} \end{aligned}$$

9. A particle executes SHM of amplitude 5 m when the particle is 3 m from its mean position, its acceleration is found to be 48 m/s^2 . Find (i) velocity (ii) time period and (iii) maximum velocity.

$$\begin{aligned} \text{9)} \quad \text{i)} \quad a &= A \omega^2 \cos(\omega t + \delta) \\ \Rightarrow a &= \omega^2 x \quad \left| \begin{array}{l} A = 5 \text{ m} \\ a = 48 \text{ m/s}^2 \\ \text{if } x = 3 \text{ m} \end{array} \right. \\ \Rightarrow \omega &= \sqrt{\frac{a}{x}} = \sqrt{\frac{48}{3}} = 4 \text{ rad/s} \\ \therefore v &= \omega \sqrt{A^2 - x^2} = 4 \sqrt{5^2 - 3^2} = 16 \text{ m/s} \\ \text{ii)} \quad T &= \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57 \text{ sec} \\ \text{iii)} \quad v_{\text{max}} &= \omega A = 4 \times 5 = 20 \text{ m/s} \end{aligned}$$

10. Particle executes harmonic motion about the point $x = 0$; at $t = 0$ it has displacement $x = 0.37 \text{ cm}$ and zero velocity. The frequency of the motion is 0.25 Hz , determine, (i) the period, (ii) the angular frequency, (iii) the amplitude, (iv) the displacement at $t = 3.0 \text{ s}$, and (v) the velocity at $t = 3.0 \text{ s}$.

$$\begin{aligned} \text{10)} \quad \text{i)} \quad T &= \frac{1}{f} = \frac{1}{0.25} = 4 \text{ sec} \\ \text{ii)} \quad \omega &= \frac{2\pi}{T} = \frac{2\pi}{4} = 1.57 \text{ rad/s} \\ \text{iii)} \quad x &= A \cos(\omega t) \\ x &= A = 3.7 \times 10^{-3} \text{ m} \quad \left| \begin{array}{l} t = 0 \text{ s} \\ v = 0 \text{ m/s} \end{array} \right. \\ \text{iv)} \quad x_{t=3} &= A \cos(\omega t) \\ &= 3.7 \times 10^{-3} \cos(1.57 \times 3) \\ &= 3.6875 \times 10^{-3} \text{ m} \\ \text{v)} \quad v &= A \omega \sin(\omega t) \\ v_{t=3} &= 3.7 \times 10^{-3} \sin(1.57 \times 3) \\ &= 3.0381 \times 10^{-4} \text{ m/s} \end{aligned}$$

11. A body oscillates with SHM according to the equation $x = 10 \cos(3\pi t + \frac{\pi}{3})$. Calculate (i) displacement at $t=2.5s$, (ii) velocity at $t=3.0s$, (iii) acceleration when $t=2s$, and (iv) maximum velocity.

12. A particle executes simple harmonic motion given by the equation $x = 10 \sin(10t - \frac{\pi}{6})$. Calculate (i) frequency, (ii) time period, (iii) the maximum displacement, (iv) the maximum velocity, and (v) the maximum acceleration.

11]

$$x = 10 \cos(3\pi t + \frac{\pi}{3})$$

$$i) x_{t=2.5} = 10 \cos(3\pi \times 2.5 + \frac{\pi}{3}) = 9.09 \text{ m}$$

$$ii) v = 30\pi \sin(3\pi t + \frac{\pi}{3})$$

$$v_{t=3} = 30\pi \sin(9\pi + \frac{\pi}{3}) = 46.1540 \text{ m/s}$$

$$iii) a = 90\pi^2 \cos(3\pi t + \frac{\pi}{3})$$

$$a_{t=2} = 90\pi^2 \cos(6\pi + \frac{\pi}{3}) = 835.2415 \text{ m/s}^2$$

$$iv) v_{max} = \omega A$$

$$= 3\pi \times 10$$

$$= 30\pi = 94.24 \text{ m/s}$$

12]

$$x = 10 \sin(10t - \frac{\pi}{6})$$

$$i) 2\pi f = 10$$

$$\Rightarrow f = \frac{10}{2\pi} = 1.5915 \text{ Hz}$$

$$ii) T = \frac{1}{f} = \frac{1}{1.5915} = 0.6283 \text{ sec}$$

$$iii) x_{max} = A = 10 \text{ m}$$

$$iv) v_{max} = A\omega = 10 \times 10 = 100 \text{ m/s}$$

$$v) a = 10^3 \sin(10t - \frac{\pi}{6})$$

$$a_{max} = 10^3 \text{ m/s}^2 \quad [\because \sin(\frac{\pi}{6}) = 1]$$

13. A mass oscillates with an amplitude of 4.00 m, a frequency of 0.5 Hz and a phase angle of $\pi/4$.

(i) What is the period T ? (ii) Write an equation for the displacement of the particle, and (iii) Calculate the velocity and acceleration of the object at time $t=5s$.

14. A 2.00 kg block is attached to a spring and force constant of the spring is $k = 196 \text{ N/m}$. The block is held a distance of 5.00 cm from equilibrium and released at $t = 0$.

(i) Find the angular frequency ω , the frequency f , and the period T .

(ii) Write an equation for x vs. time.

13] i) $T = \frac{1}{f} = \frac{1}{0.5} = 2 \text{ sec}$

ii) $x = 4 \cos(\pi t + \frac{\pi}{4})$

iii) $v = 4\pi \sin(\pi t + \frac{\pi}{4})$

$$a = 4\pi^2 \cos(\pi t + \frac{\pi}{4})$$

$$v_{t=5} = 4\pi \sin(5\pi + \frac{\pi}{4}) = 8.88 \text{ m/s}$$

$$v_{t=5} = 4\pi \sin(5\pi + \frac{\pi}{4}) = 8.88 \text{ m/s}$$

$$a_{t=5} = 4\pi^2 \cos(5\pi + \frac{\pi}{4}) = 27.91 \text{ m/s}^2$$

14]

i) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 7\sqrt{2} \text{ rad/s}$

$$f = \frac{\omega}{2\pi} = \frac{7\sqrt{2}}{2\pi} = 1.57 \text{ Hz}$$

$$T = \frac{1}{f} = \frac{1}{1.57} = 0.6346 \text{ sec}$$

ii) $x = 0.05 \cos(7\sqrt{2}t + \delta)$

15. A body of mass 500 gm is suspended from a spring of negligible mass and it stretches the spring by 7 cm. For a displacement of 3 cm it is given a downward velocity 40 cm/s. Calculate (i) the spring constant, (ii) the angular frequency, (iii) the time period (iv) the initial potential energy, (v) the initial kinetic energy, and (vi) the amplitude of the ensuing motion of the spring.

15]

i) $v = \omega \sqrt{A^2 - x^2} = 2\sqrt{10} \text{ rad/s}$

$\Rightarrow \omega = \frac{.4}{\sqrt{(.07)^2 - (.03)^2}} = 2\sqrt{10} \text{ rad/s}$

$k = \omega^2 m = (2\sqrt{10})^2 \times .5 = 20 \text{ N/m}$

ii) $\omega = 2\sqrt{10} \text{ rad/s}$

iii) $T = \frac{2\pi}{\omega} = .9934 \text{ sec}$

iv) $E_p = \frac{1}{2} k x^2 = \frac{1}{2} \times 20 \times (.07)^2 = .049 \text{ J}$ [initial]

v) $E_k = 0 \text{ J}$ [initial]

vi) $A = .07 \text{ m}$

$m = .5 \text{ kg}$
 $A = .07 \text{ m}$
 $x = .03 \text{ m}$
 $v = .4 \text{ m/s}$

17. Suppose a spring block-system moves between top and bottom point of a tall buildings as a mass dampers. The block has mass $m = 2.72 \times 10^5 \text{ kg}$ and designed to oscillate at a frequency $f = 10 \text{ Hz}$ with amplitude $x_m = 20 \text{ cm}$. Calculate (i) the mechanical energy, (ii) what is the block speed as it passes through the equilibrium point?

17]

i) $E = \frac{1}{2} k A^2$

$= \frac{1}{2} \omega^2 m A^2$

$= \frac{1}{2} (2\pi \times 10)^2 \times 2.72 \times 10^5 \times (.2)^2$

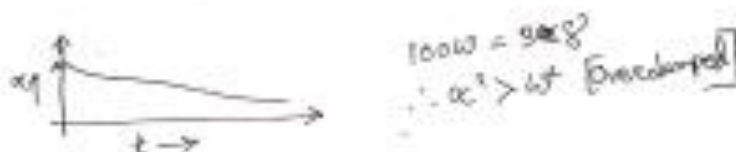
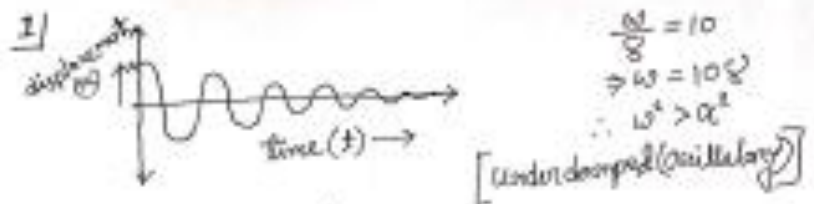
$= 21.47 \times 10^6 \text{ J}$

ii) $v_{\text{max}} = \omega A$

$= 2\pi \times 10 \times .2$

$= 4\pi = 12.56 \text{ m/s}$

1. Draw displacement vs time graph for $\omega/\gamma = 10$, $\omega/\gamma = 0.5$, and $\omega/\gamma = 0.03$.



2. In oscillatory circuit $L = 0.4\text{h}$, $C = 0.0020\mu\text{F}$. (i) What is maximum value of resistance (R) for the circuit to be oscillatory? and (ii) What is its resonant frequency?
3. Find whether the discharge of capacitor through the following inductive series circuit is oscillatory or not. Given, $C = 0.1\mu\text{F}$, $L = 10\text{mh}$, and $R = 200\Omega$. If oscillatory, find the frequency of oscillation and resonant frequency. If it is parallel circuit, then find out the similar characteristics of that circuit.

2/

i) $\left(\frac{R_{\max}}{2L}\right)^2 = \frac{1}{LC}$
 $\Rightarrow R_{\max} = \sqrt{\frac{4L}{C}} = \sqrt{\frac{4 \times 0.4}{0.0020 \times 10^{-6}}} = 28284.27\Omega$

ii) $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{0.4 \times 0.0020 \times 10^{-6}}} = 5626.97\text{Hz}$

$L = 0.4\text{h}$
 $C = 0.0020 \times 10^{-6}\text{F}$

3/

$\alpha^2 = \frac{R^2}{4L^2} = \frac{(200)^2}{4 \times (10 \times 10^{-3})^2} = 100 \times 10^6$
 $\omega^2 = \frac{1}{LC} = \frac{1}{10 \times 10^{-3} \times 0.1 \times 10^{-6}} = 10^9$
 $\therefore \omega^2 > \alpha^2$ [oscillatory]

for series,
 $\omega_d = \sqrt{\omega^2 - \alpha^2}$
 $= \sqrt{10^9 - 10^8}$
 $= 30000\text{ rad/s}$
 Resonant frequency
 $f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{10^{-3} \times 0.1 \times 10^{-6}}} = 5032.92\text{Hz}$
 $b = 0.7\text{kg/s}$

for parallel
 $\omega_d = \sqrt{\omega^2 - \alpha^2}$
 $= \sqrt{10^9 - \frac{1}{4 \times 10^{-3} \times 0.1 \times 10^{-6}}}$
 $= 19364.91\text{ rad/s}$

4. For a damped oscillator $m = 250\text{g}$, $k = 85\text{N/m}$ and $b = 70\text{gm/s}$. (i) What is the period of the motion? (ii) How long does it take for the amplitude of the damped oscillations to drop to half its initial value? (iii) How many oscillations does it complete in life time? (iv) What is its life time? (v) The maximum displacement of undamped oscillator is 35 cm . If the damping is stopped after 20 cycles, What is the damping energy? (vi) What is the ratio of the oscillation amplitude to the initial oscillation amplitude at this cycle?

$\underline{\text{4}} \quad m = 0.25\text{kg} \quad k = 85\text{N/m} \quad b = 0.07\text{kg/s}$
 i) $T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{0.25}{85}} = 1.34\text{s}$
 ii) $A_0 e^{-\alpha t} = \frac{A_0}{2}$
 $\Rightarrow e^{-\alpha t} = \frac{1}{2}$
 $\Rightarrow -\alpha t = \ln\left(\frac{1}{2}\right)$
 $\Rightarrow t = \frac{\ln(2)}{-\alpha}$
 $\quad \quad \quad = \frac{\ln(2)}{-\left(\frac{b}{2m}\right)}$
 $\quad \quad \quad = \frac{\ln(2)}{-\left(\frac{0.07}{2 \times 0.25}\right)}$
 $\quad \quad \quad = 4.95\text{s}$
 iii) number of oscillations
 $n = \frac{t}{T} = \frac{4.95}{1.34} = 3.69$
 iv) $\gamma = \frac{1}{\delta} = \frac{m}{b} = \frac{0.25}{0.07} = 3.57$
 v) $E = E_0 e^{-\alpha t}$ $\left[t = T \times 20 = 6.8\text{s} \right]$
 $= \frac{1}{2} k A^2 e^{-\alpha t}$
 $= \frac{1}{2} \times 85 \times (0.35)^2 e^{-\frac{0.07}{0.5} \times 6.8}$
 $= 0.77\text{J}$
 vi) $\frac{A}{A_0} = \frac{A_0 e^{-\alpha t}}{A_0} = e^{-\alpha t}$
 $= e^{-\frac{b}{2m} t}$

5. The equation of a travelling wave is $y = 4.0 \sin(0.10x - 2t)$. Find (i) amplitude, (ii) wavelength, (iii) speed, and (iv) frequency of wave.
6. At time $t=0$ the displacement of a particle in a medium is $y = 4.0 \sin 2\pi\left(\frac{x}{100}\right)$ and the velocity of wave is 30cm/s . Find the displacement equation when $t = 3\text{s}$.

$\underline{5} \quad y = 4 \sin(0.10x - 2t)$
 $= 4 \sin(0.10(20\lambda - x))$
 compare with $y = A \sin \frac{2\pi}{\lambda}(v\lambda - x)$
 i) $A = 4\text{m}$
 ii) $0.10 = \frac{2\pi}{\lambda}$
 $\Rightarrow \lambda = 62.83\text{m}$
 iii) $v = 20\text{m/s}$
 iv) $v = f\lambda \Rightarrow f = \frac{v}{\lambda} = \frac{20}{62.83}$
 $= 0.31\text{Hz}$

$\underline{6} \quad y = 4 \sin 2\pi\left(\frac{x}{100}\right) \quad [t=0]$
 $\therefore y = 4 \sin \frac{2\pi}{100}(30t + x)$ general form
 $y_{t=3} = 4 \sin \frac{2\pi}{100}(90 + x)$
 $\quad \quad \quad = 4 \sin \frac{2\pi}{100}(90 + x)$

7. When a simple harmonic wave is propagated through a medium, the displacement of the particle at any instant of time is given by $y = 5.0 \sin \pi(360t - 0.15x)$. Calculate (i) the amplitude of the vibrating particle, (ii) wave velocity, (iii) wave length, (iv) frequency and (v) time period.
8. A simple harmonic wave of amplitude 8 units travels a line of particles in the direction of positive X axis. At any instant for a particle at a distance of 10 cm from the origin, the displacement is +6 units and at a distance a particle from the origin is 25 units, the displacement is +4 units. Calculate the wavelength.

Q7

$$y = 5 \sin \pi(360t - 0.15x)$$

$$= 5 \sin \pi \cdot 15 \pi (2400t - x)$$

Compare with $y = A \sin \frac{2\pi}{\lambda}(vt - x)$

i) $A = 5 \text{ m}$

ii) $V = 2400 \text{ m/s}$

iii) $\frac{2\pi}{\lambda} = 0.15\pi$
 $\Rightarrow \lambda = 13.33 \text{ m}$

iv) $v = f\lambda$
 $\Rightarrow f = \frac{v}{\lambda} = \frac{2400}{13.33} = 180 \text{ Hz}$

v) $T = \frac{1}{f} = 5.55 \times 10^{-3} \text{ s}$

Q8

$$y = A \sin \frac{2\pi}{\lambda}(vt - x)$$

$$\Rightarrow y = A \sin 2\pi \left(\frac{v}{\lambda}t - \frac{x}{\lambda} \right)$$

$$\Rightarrow \frac{y}{A} = \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right)$$

$v = f\lambda$
 $f = \frac{1}{T}$

Now,

$$\frac{6}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{10}{\lambda} \right) \quad \text{--- (i)}$$

$$\frac{4}{8} = \sin 2\pi \left(\frac{t}{T} - \frac{25}{\lambda} \right) \quad \text{--- (ii)}$$

$$\text{(i)} \rightarrow \frac{t}{T} - \frac{10}{\lambda} = 0.134$$

$$\text{(ii)} \rightarrow \frac{t}{T} - \frac{25}{\lambda} = 0.083$$

$$\text{So, } 0.134 + \frac{10}{\lambda} = 0.083 + \frac{25}{\lambda}$$

$$\Rightarrow \lambda = \frac{15}{0.051} = 294.11 \text{ cm}$$

$f = \frac{1}{T}$

9. Determine the length and frequency of a simple pendulum that will swing back and forth in simple harmonic motion with a period of 5.00 s.
10. Find out the resultant amplitude, node and antinode points in terms of λ of the following equations:
 $y_1 = A \cos(\frac{1}{3}kx + \omega t)$ and $y_2 = A \cos(\frac{1}{3}kx - \omega t)$.

$$\begin{aligned}
 9) \quad T &= 2\pi \sqrt{\frac{L}{g}} \\
 &\Rightarrow 2\pi \sqrt{\frac{L}{g}} = 5 \\
 &\Rightarrow L = \left(\frac{5}{2\pi}\right)^2 \times 9.8 \\
 &= \boxed{6.21 \text{ m}}
 \end{aligned}$$

$$\begin{aligned}
 f &= \frac{1}{T} \\
 &= \frac{1}{5} = 0.2 \text{ Hz}
 \end{aligned}$$

$$\begin{aligned}
 10) \quad y_1 &= A \cos(\frac{1}{3}kx + \omega t) \quad \text{--- (I)} \\
 y_2 &= A \cos(\frac{1}{3}kx - \omega t) \quad \text{--- (II)} \\
 y &= y_1 + y_2 \\
 &= A \cos(\frac{1}{3}kx + \omega t) + A \cos(\frac{1}{3}kx - \omega t) \\
 &= A \left[\cos(\frac{1}{3}kx + \omega t) + \cos(\frac{1}{3}kx - \omega t) \right] \\
 &= A \left[2 \cos \frac{kx}{3} \cos \omega t \right] \quad \left[\because 2 \cos \frac{kx}{3} = \text{constant} \right] \\
 &= 2A \cos \frac{kx}{3} \cos \omega t \quad \left[\text{Here } A_0 = 2A \cos \frac{kx}{3} \right] \\
 &= A_0 \cos \omega t \quad \left[\text{which is resultant amplitude} \right]
 \end{aligned}$$

Node point:

$$\begin{aligned}
 \cos \frac{kx}{3} &= 0 \\
 \Rightarrow \cos \frac{kx}{3} &= \cos(\frac{\pi}{2}), \cos(\frac{3\pi}{2}), \cos(\frac{5\pi}{2}), \dots \\
 \Rightarrow \frac{kx}{3} &= \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \dots \\
 \Rightarrow \frac{kx}{3} &= \frac{\pi}{2}(2n+1) \quad \left[\text{here } n=0, 1, 2, 3, \dots \right] \\
 \Rightarrow x &= \frac{3}{2} \frac{\pi}{k} (2n+1) \\
 &= \frac{3}{2} \frac{\lambda}{2\pi} (2n+1) \\
 &= \frac{3}{4} \lambda (2n+1)
 \end{aligned}$$

Antinode:

$$\begin{aligned}
 \cos(\frac{kx}{3}) &= \pm 1 \\
 \Rightarrow \cos(\frac{kx}{3}) &= \cos(0), \cos(\pi), \cos(2\pi), \cos(3\pi), \dots \\
 \Rightarrow \frac{kx}{3} &= 0, \pi, 2\pi, 3\pi \\
 \Rightarrow \frac{kx}{3} &= n\pi \quad \left[\text{here } n=0, 1, 2, 3, \dots \right] \\
 \Rightarrow x &= \frac{3}{2\pi} n\pi = \frac{3}{2} n\lambda
 \end{aligned}$$

11. Two cosine waves have phase velocity $C_1 = 2$ cm/s, $C_2 = 3$ cm/s, and corresponding wavelength $\lambda_1 = 4$ cm, $\lambda_2 = 3$ cm. Find out the angular frequency and group velocity.

15. Suppose the displacement of a DHM is expressed as $x = Ae^{-\alpha t} \cos(\omega_d t + \delta)$. Show that total energy of DHM $E = E_0 e^{-2\alpha t}$.

⑪

$$k_1 = \frac{2\pi}{\lambda_1} = \frac{2\pi}{4} \text{ cm}^{-1} \quad c_1 = 2 \text{ cm/s}$$

$$k_2 = \frac{2\pi}{\lambda_2} = \frac{2\pi}{3} \text{ cm}^{-1} \quad c_2 = 3 \text{ cm/s}$$

Angular frequency,

$$\omega_1 = c_1 k_1 = 2 \times \frac{2\pi}{4} = \pi \text{ rad/s}$$

$$\omega_2 = c_2 k_2 = 3 \times \frac{2\pi}{3} = 2\pi \text{ rad/s}$$

group velocity

$$v_g = \frac{d\omega}{dk} = \frac{\omega_2 - \omega_1}{k_2 - k_1} = \frac{2\pi - \pi}{\frac{2\pi}{3} - \frac{2\pi}{4}} = 6 \text{ cm/s}$$

15]

$$x = A e^{-\alpha t} \cos(\omega_d t + \delta)$$

$$= A_0 \cos(\omega_d t + \delta) \quad \left[\text{here } A_0 = A e^{-\alpha t} \right]$$

$$\therefore E_d = \frac{1}{2} k A_0^2$$

$$= \frac{1}{2} k (A e^{-\alpha t})^2$$

$$= \frac{1}{2} k A^2 e^{-2\alpha t}$$

$$= \frac{1}{2} E_0 e^{-2\alpha t}$$

SHM Energy

$$E_0 = \frac{1}{2} k A^2$$